

MATCHMINDAI

Real-Time Sports Intelligence & 10,000-Run Monte Carlo Simulation Engine

Live Web App URL: matchmindai.streamlit.app	GitHub Repo: github.com/Soham-ss/MatchMindAI
Core Tech Stack: Python, Streamlit, Scikit-Learn, Plotly	Engine: 10,000 Monte Carlo Runs + ML RandomForest

■ Slide 2: The Problem Statement

■ Problem 1: Static Historical Data Only	Traditional prediction tools only look at static past averages. They ignore today's live pitch reports, weather changes, and stadium dew factors.
■ Problem 2: Messy AI Data Dumps	General AI chatbots return messy raw metadata links (e.g. Source [1]...) instead of simple, crisp 1-line direct answers.
■ Problem 3: Lack of Match Simulation	Cricket fans and fantasy players cannot simulate thousands of ball-by-ball match scenarios to view score distribution ranges.

■ Slide 3: Our Solution (MatchMindAi)

■ 10,000 Monte Carlo Runs	Simulates 10,000 ball-by-ball matches per click based on team strength ratings, venue pitch profiles, and stadium dew factors.
■ 1-Line Factual Answers	Instantly answers player team queries, Orange Cap winners, Purple Cap winners, and Champions for any year (2008–2026).
■ Live Web Intelligence	Scrapes live Google News RSS feeds to incorporate today's pitch, weather, and team squad updates.
■ High-Energy Cyber UI	Dark cyberpunk theme, live stadium video header, cursor depth particle radar, and sticky bottom Chat Box.

■ Slide 4: Main Platform Features

	Recognizes user name introductions personally (e.g. <i>'Hi my name is Soham'</i> → <i>'Hello Soham! ■'</i>).	■ 18-Year IPL Database	Maps (e.g.
	Detects invalid pre-2008 queries (<i>'who won in 2005'</i>) explaining IPL started in 2008 with Shaun Marsh.	■ 1-Click Match Predictor	Pick score

Slide 5: System Architecture & Workflow

Step 1: User Input	User inputs a question or selects 2 teams & stadium venue.
Step 2: Intent Router	Classifies query into Greeting, Bot Identity, IPL History Lookup, or Match Prediction.
Step 3: Web Scraper Agent	Scrapes live Google News RSS feeds for real-time weather & squad updates.
Step 4: Monte Carlo Agent	Executes 10,000 ball-by-ball stochastic match outcome simulations.
Step 5: ML RandomForest Agent	Queries pre-trained machine learning model on historical IPL match features.
Step 6: Blended Ensemble	Combines 55% Monte Carlo + 45% ML to compute final win probabilities and report.

■ Slide 6: Machine Learning & Monte Carlo Engine

■ RandomForest Classifier	Trained on historical IPL datasets using scikit-learn. Encodes teams, venue, toss winner, and toss decision. Pre-compiled weights are saved as .pkl binary files in models/.
■ Monte Carlo Math	Calculates run probabilities per ball based on team rating (CSK 88, MI 87, RCB 85) and venue dew factor (Wankhede 0.12). Runs 10,000 iterations in milliseconds.
■ Score Distributions	Generates interactive Plotly bar charts showing expected score ranges and average totals for both competing teams.

■ Slide 7: Technical Challenges Faced & Solved

■ GitHub 100 MB Limit Error	■ Solution: Removed IPL.csv (104 MB) from git history because model weights were already compiled into .pkl binary files in models/.
■ Streamlit Cloud Path Errors	■ Solution: Replaced relative string paths with Python's os.path.abspath to dynamically locate exact folder paths on Linux.
■ Missing Cloud Packages	■ Solution: Added beautifulsoup4 and python-dotenv to requirements.txt and wrapped imports in safe try...except blocks.
■ Phrasing Variations	■ Solution: Expanded regex keywords to catch all tense variations ('who wins', 'wins', 'win', 'won').

■ Slide 8: UI/UX & High-Energy Design System

■ Cyberpunk Dark Aesthetic	Designed with a sleek #030712 background and HSL tailored emerald accents (#10b981, #34d399).
■ Modern Typography	Uses Google Font (Outfit) for maximum legibility across laptops, tablets, and mobile devices.
■ Dynamic Visual FX	Features a looping HD stadium lights video banner and an interactive 4D particle cursor depth radar.
■ Sticky Bottom Chat Input	Pinned Chat Box stays permanently visible at the bottom of the viewport when scrolling long chat histories.

■ Slide 9: Cloud Deployment & Infrastructure

■ Live Hosting	Hosted live on Streamlit Community Cloud with 100% uptime. Accessible globally at matchmindai.streamlit.app .
■ Source Control	Version-controlled on GitHub (github.com/Soham-ss/MatchMindAI).
■ Automated CI/CD	Pushing code updates to GitHub main branch automatically triggers a rebuild on Streamlit Cloud within ~30 seconds.

■ Slide 10: Conclusion & Future Roadmap

■ Conclusion	MatchMindAi successfully combines Artificial Intelligence, Machine Learning, Monte Carlo Simulations, and Real-Time Internet Scraping into a fast, accurate, and visually stunning web platform.
■ Live Score API Integration	Connect live cricket APIs for real-time win probability shifts during active live matches.
■ Fantasy Team Optimizer	Auto-generate optimal Dream11 / My11Circle lineups based on Monte Carlo expected points.
■ Multi-Sport Expansion	Extend simulation models to Football (Premier League, Champions League) and Basketball (NBA).